Using Embedding Models and a Vector Database to look at similarity between code.

The focus of this project will be to use previously trained embedding models, such as those by Ollama, OpenAI and Google, to build a large vector database containing the vector representation of code. This database will provide various applications such as an API which a user could query which a chunk of input code that returns a list of similar matching code chunks.

Embedding models are designed to show representations of values of objects in mathematical vector form, in this project the focus will be on embedding of chunks of code. The vector representation of such objects then allows comparison between similar objects using the properties of the vector. In this project the similarity of vectors will be measured using cosine similarity, in which the angle between vectors is used as a measure of similarity. This can be achieved through the use of a vector database.

Additionally, as VS Code plugin may be created which will allow users to query the API in real-time as they write code which could enhance developer productivity.

Overall, the main focus of this project would be to to look at different embedding models and different ways of chunking code to build a database that contains information about code similarity using historical data, e.g. by taking open source code from platforms such as GitHub.